Assignment 6

Learning goals:

Apply the concepts and methods presented in chapter “Symbol Table” of the lecture to some exercise problems.

Presentation:

Please prepare the solutions for informal presentation in class, and fill in the online check list in ZEUS.

Task 1 Symbol Table with primitive types

Consider the following MJ-code snippet.

```
program P
final int a = 5;
int b, c;
{
    void x (int c, char d)
        { }
    void main(int f)
        int g;
        { }
}
```

Show the structure of the symbol table after parsing the opening brace of the main method.

Task 2 Symbol Table with arrays

Consider the following MJ-code snippet.

```
program P
int[] a;
int[] b;
{
void main()
    char[] c, d;
    { }
}
```

Show the structure of the symbol table after parsing the opening brace of the main method.

Task 3 Symbol Table with classes

Consider the following code snippet.

```
program P
class A { char[] f; }
class B { int x, y; }
{
   void main()
   A v, w; B x, y;
   {
   }
}
```

Show the structure of the symbol table after parsing the opening brace of method main().

Task 4 MJ symbol table

Extend the MicroJava compiler to output the symbol table using the MJ.SymTab package. Add a --symdump argument to enable symbol table output. Create a method 'printSymTabPretty' which allows the compiler to print information of variables and constants that are accessible from the scope of the main method just before it terminates. For variables, print type, level and address. For constants, print type, level and value. For classes, print type, level and address as well as their fields.

Example-Input – any MJ-compliant file.

```
program P
final char a = 'a';
int b;
int[] c;
class A { char[] f; }
class B { int g; char h; }
{
   void main()
   A v; B x;
   {  }
   void p()
   A q;
   {  }
}
```

Example-Output - format is freely customizable, but must be comprehensible without explanation:

```
Var: v        Type: Class    Level: 1    Address: 0
- Field:f     Type: Arr      Level: 1    Address: 0
Var: x        Type: Class    Level: 1    Address: 1
- Field:g     Type: Int      Level: 1    Address: 0
- Field:h     Type: Char     Level: 1    Address: 1
Con: a        Type: Char     Level: 0    Value: 97
Var: b        Type: Int      Level: 0    Address: 0
Var: c        Type: Arr      Level: 0    Address: 1
```